Uncovering Local Consumption Dynamics during COVID-19: Evidence from a Spatial Dynamic Factor Model

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Abstract

This paper develops a spatial dynamic factor model to study how household consumption evolved across regions during the COVID-19 pandemic. By incorporating multiple latent factors and spatial dependence, the model captures both temporal persistence and geographic spillovers in consumption behavior. Using weekly data, we find that Economic Impact Payments had diminishing effects over time, with rural areas showing stronger consumption responses than urban and suburban counterparts. Suburban counties, in contrast, displayed larger unexplained consumption fluctuations, a pattern that may reflect pandemic-related migration and labor market reallocation. Monte Carlo simulations show traditional models are biased, underscoring the need for spatial approaches. Results stress the importance of geographically targeted policy interventions.

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1 Introduction

The COVID-19 pandemic caused large and uneven disruptions to economic activity across regions. Understanding how these shocks affected household consumption is central to assessing their economic impact. While a growing literature has documented aggregate changes in spending, less is known about how consumption responses propagate across geographically connected areas and how local spillovers shape the spatial pattern of recovery.

A large body of literature, including Blundell and Preston (1998) and Krueger and Perri (2006), has examined how income shocks and credit constraints influence consumption, typically treating households or regions as independent units. **This approach overlooks the fact that economic activity is inherently interconnected across space.** Consumption patterns in one region can affect neighboring areas through labor mobility, supply chains, and fiscal linkages, giving rise to cross-regional spillovers that standard aggregate or local models fail to capture.

To address these limitations, we develop a spatial dynamic factor framework that captures how consumption shocks propagate across regions. The model extends standard dynamic factor approaches by introducing spatial dependence directly in the factor loadings, allowing regions to differ in their exposure to common shocks in a way that reflects geographic linkages. This structure captures cross-regional spillovers while remaining computationally tractable for large panels by exploiting the banded precision matrix implied by the first-order Markovian factor dynamics. Monte Carlo evidence shows that ignoring this spatial structure leads to biased inference.

Using county-level data from the Opportunity Insights Economic Tracker, **we examine consumption patterns across 1,679 U.S. counties over 154 weekly observations**

beginning in the third week of January 2020 and ending in December 2022, incorporating key economic, pandemic-related, and policy variables. We find substantial spatial heterogeneity in consumption responses: rural areas exhibit more predictable trajectories while suburban counties show larger unexplained deviations that may reflect pandemic-related migration and labor market reallocation. Stimulus payments had limited short-term effects in urban and suburban areas, where households allocated funds toward savings or debt repayment, but generated more immediate consumption gains in rural regions. This rural boost diminished across subsequent payment rounds, suggesting declining marginal propensities to consume and pointing to the importance of geographically targeted fiscal policy. Consumer spending was more strongly influenced by perceived health risks than by formal lockdowns or income changes, consistent with evidence that voluntary behavior drove much of the economic contraction.

This paper makes two contributions. First, it develops a spatial dynamic factor model with a computationally efficient simulation smoother that scales to large geographic panels. Second, it provides new county-level evidence on spatially heterogeneous consumption responses to the COVID-19 pandemic, with direct implications for the targeting of stabilisation policies.

The remainder of the paper is organised as follows. Section 2 reviews the related literature on consumption dynamics and spatial dynamic factor models. Section 3 develops the spatial dynamic factor model, covering the economic foundations, the factor dynamics, and the spatial loading structure. Section 4 presents the Gibbs sampler that exploits the banded precision matrix for efficient Cholesky-based simulation smoothing, with prior distributions and likelihood function detailed alongside. Section 5 provides Monte Carlo evidence on the performance of the proposed estimator relative to the Kalman filter. Section 6

introduces the county-level dataset, which is then used in Section 7 to study consumption dynamics during the COVID-19 pandemic. Section 8 concludes.

2 Literature Review

Recent work has documented large shifts in consumer spending during the COVID-19 pandemic, driven by demographic differences, policy responses, and evolving health risks. Baker et al. (2020) and Chetty et al. (2024) showed that although consumption fell across all households, declines were larger and more persistent among higher-income groups. Sectoral patterns were also uneven: Alexander and Karger (2023) reported early surges in grocery purchases and sharp contractions in discretionary categories. Beyond consumption, heterogeneous regional adjustments in real estate markets were identified by Heiniger et al. (2025), underscoring the broader spatially uneven effects of the pandemic.

A parallel literature examines the determinants of these fluctuations. While Coibion et al. (2025) attributed early spending declines to lockdowns, Goolsbee and Syverson (2021) emphasized the role of voluntary behavioral responses. Government support played a key role: Parker et al. (2022) and Coibion et al. (2025) found that Economic Impact Payments generated rapid spending increases, especially among lower-income households.

Vaccination campaigns also contributed to stabilization. Deb et al. (2021) showed that rising vaccination rates reduced economic uncertainty and supported recovery. Methodologically, high-frequency tracking tools such as the dynamic factor models used by Lewis et al. (2021) have improved our understanding of real-time economic conditions, though primarily at the national level.

From a methodological perspective, our paper builds on the literature on dynamic factor models with spatial structure. Lopes et al. (2008) introduced spatial

dynamic factor analysis using a Kalman filter approach, modelling spatial dependence in the loadings via a Matérn covariance function; however, their method scales poorly to large cross-sections due to the cost of repeated $n \times n$ matrix inversions in the filter recursions. Han and Lee (2016) developed Bayesian spatial panel models incorporating common factors and random coefficients, where spatial dependence is introduced through a conditional autoregressive prior on the idiosyncratic errors rather than the loadings, and does not separately identify the spatial transmission mechanism through shock sensitivity. Qin et al. (2025) recently proposed Bayesian dynamic matrix factor models for matrix-valued time series, where dimensionality reduction proceeds simultaneously along both the row and column dimensions; their framework is designed for settings where the observable data have a natural matrix organisation (such as assets $\times$ characteristics), and cross-sectional dependence is absorbed through the Kronecker structure of the loadings rather than modelled as a spatial process tied to geographic proximity. Frühwirth-Schnatter et al. (2025) provide the most complete recent treatment of Bayesian factor model identification and sparsity, developing generalised lower triangular (GLT) identification conditions where factor loadings are treated as independent across cross-sectional units with no spatial or geographic structure; our identification strategy follows instead the classical lower triangular normalisation of Geweke and Zhou (1996) and Lopes and West (2004), which imposes an identity matrix in the first p rows of Λ (see online supplement Section D.1).

Our contribution relative to these frameworks is not to propose an entirely new model class, but to deliver two specific advances suited to a large-n geographic panel. First, we introduce spatial autoregressive structure directly into

the factor loadings via the weight matrix W, which explicitly encodes geographic proximity and identifies the spatial transmission mechanism separately from the temporal dynamics (LeSage and Pace, 2009). Second, we exploit the resulting banded precision matrix to achieve computationally efficient MCMC inference via Cholesky-based simulation smoothing, making the approach scalable to large geographic panels; Monte Carlo evidence confirms that ignoring spatial structure in the loadings leads to biased inference.

3 A Spatial Dynamic Factor Model of Consumption

This section builds the spatial dynamic factor model from its theoretical underpinnings. We begin with consumption theory under uncertainty, which motivates the factor structure, before introducing the spatial autoregressive loading specification that captures cross-regional spillovers.

Traditional consumption models based on the Permanent Income Hypothesis (PIH) posit that rational consumers smooth consumption over time, responding only to unexpected changes in permanent income (Hall, 1978). However, empirical evidence suggests that consumption growth is not solely determined by income shocks but is also influenced by precautionary saving motives and income uncertainty (Carroll, 1997). These findings highlight the need for a theoretical framework that explicitly accounts for both uncertainty and macroeconomic risk factors to accurately capture consumption dynamics.

To address these limitations, this framework integrates both the precautionary saving hypothesis and the presence of liquidity-constrained consumers who deviate from intertemporal optimization. The model extends the standard buffer-stock saving approach developed by Carroll (1997), where households maximize expected lifetime utility while facing income uncertainty. The household's optimization problem is given by:

$$\max_{\{c_t\}_{t=0}^{\infty}} E_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right], \tag{1}$$

subject to the intertemporal budget constraint $A_{t+1} = R_{t+1}\big(A_t + y_t - c_t\big)$. Here, c_t represents consumption, y_t denotes income, and A_t is the stock of assets, while R_t represents the gross return on savings, and β is the discount factor.

Under constant relative risk aversion (CRRA) preferences, the household's utility function is specified as:

$$u(c) = \frac{c^{1-\theta}}{1-\theta}, \quad \theta > 0. \tag{2}$$

The first-order Euler equation for optimal consumption then takes the form:

$$c_t^{-\theta} = \beta R_{t+1} E_t \left[c_{t+1}^{-\theta} \right]. \tag{3}$$

As detailed in the online supplement Section A, taking logarithms and applying the moment-generating function (MGF) property of a normally distributed variable, the log-linearized consumption growth equation is obtained:

$$E_t \left[\Delta \ln c_{t+1} \right] = \frac{1}{\theta} \left(r_{t+1} - \tau \right) + \frac{\theta}{2} \mathrm{Var}_t \left(\Delta \ln c_{t+1} \right), \tag{4}$$

where $r_{t+1} = \ln R_{t+1}$ and $\tau = -\ln \beta$ is the rate of time preference (derived in the online supplement Section A). This expression demonstrates that expected consumption growth depends not only on the intertemporal substitution effects of interest rates but also on precautionary saving responses to uncertainty, captured by the conditional variance of $\Delta \ln c_{t+1}$ given information at time t. When future income is uncertain, households increase precautionary savings, leading to higher expected consumption growth (Carroll, 1997).

3.1 Economic Foundations

The standard PIH framework assumes that all consumers engage in intertemporal optimization. However, Campbell and Mankiw (1990) challenge this assumption by introducing a two-group model in which a fraction of consumers, termed "rule-of-thumb" consumers, do not smooth consumption over time but instead consume a fixed fraction of their current income. In this framework, aggregate consumption growth is given by $\Delta c_t = \nu \Delta y_t + \varepsilon_t$, where ν represents the fraction of income that is immediately consumed. Empirical findings indicate that a significant portion of households exhibit rule-of-thumb behavior, contradicting the predictions of the standard PIH **(Campbell and Mankiw, 1990; Parker, 1999).**

The rule-of-thumb behavior has major implications for consumption dynamics. It suggests that both transitory and permanent income shocks can have immediate effects on spending, in contrast to the PIH, which posits that only changes in lifetime wealth should affect consumption. The presence of liquidity-constrained consumers strengthens the argument for explicitly incorporating income fluctuations into models of consumption growth.

3.2 Dynamic Factor Structure

Given the multiple macroeconomic factors influencing consumption, it is necessary to employ a model that accounts for both observed and unobserved sources of risk. A dynamic factor loading model provides a flexible approach by decomposing consumption growth into systematic economic influences and latent macroeconomic fluctuations. Rather than imposing an exogenous variance term, this framework allows the variance in consumption growth to emerge endogenously from the structure of the dynamic factor loadings.

Instead of explicitly including $\text{Var}_t(\Delta \ln c_{t+1})$ as a separate regressor, this framework assumes that it is captured through the variance of time-varying factor loadings and unobserved economic shocks. Specifically, the conditional variance of consumption growth is modeled as:

$$\mathrm{Var}_t\left(\Delta \ln c_{t+1}\right) = \mathrm{Var}(\Lambda F_t). \tag{5}$$

Here, Λ represents spatially dependent factor loadings that capture economic interdependencies across regions, while F_t denotes unobserved factors that evolve over time. The interaction between these components allows uncertainty in consumption growth to be determined dynamically rather than being imposed exogenously.

3.3 Model Specification

To operationalize this framework empirically, we analyse the weekly consumption behavior of individuals aggregated at the county level. For each county $i = 1, \ldots, n$ at time $t = 1, \ldots, T$, the difference in log consumption is modelled as:

$$\begin{aligned} \Delta \ln c_{it} &= \alpha + \nu \Delta \ln y_{it} + \gamma' X_{it} + \Lambda_i' F_t + \epsilon_{it}, & (6) \\ F_t &= \Phi F_{t-1} + \xi_t & (7) \end{aligned}$$

where $\Delta \ln c_{it}$ and $\Delta \ln y_{it}$ represent the growth rates of consumption and income for county i at time t, X_{it} is a k-dimensional vector of explanatory variables, $\Lambda_i = (\lambda_{i,1}, \ldots, \lambda_{i,p})'$ is the p-dimensional factor loading vector capturing spatial dependencies, $F_t = (F_{t,1}, \ldots, F_{t,p})'$ is the p-dimensional time-varying factor vector capturing unobserved consumption dynamics, $\Phi = \mathrm{diag}(\phi_1, \ldots, \phi_p)$ governs factor persistence, $\xi_t \sim N(0, \Sigma_\xi)$ with $\Sigma_\xi = \mathrm{diag}(\sigma^2_{\xi,1}, \ldots, \sigma^2_{\xi,p})$, and $\epsilon_{it} \sim N(0, \sigma_i^2)$ represents idiosyncratic shocks. The parameter $\gamma = (\gamma_1, \ldots, \gamma_k)'$ captures the marginal effects of the explanatory variables on consumption growth.

Two parameter restrictions are adopted for tractability rather than economic reasons. First, diagonal Φ preserves a sparse block structure that enables efficient estimation, as detailed in Section 4; this sparsity in the transition dynamics connects to a broader literature on sparse hidden Markov and state-space mod-

els (e.g., Zucchini and MacDonald, 2009; Koopman and Durbin, 2000), where restricting off-diagonal elements reduces the parameter space and aids identification in high-dimensional settings, though the connection is one of structural analogy rather than formal equivalence since our factors are continuous rather than discrete latent states. Second, diagonal Σ_ξ reduces the innovation parameter space from p^2 to p variance parameters; since the factors are latent constructs, uncorrelated innovations carry no specific economic content. The economic implication of both restrictions is that each factor follows an independent AR(1) with no cross-factor transmission; full Φ and Σ_ξ are natural generalisations for future work.

3.4 Spatial Dynamic Factor Loading Structure

To capture regional interdependencies in consumption uncertainty, a spatial autoregressive structure is introduced in each of the $l = (1, \ldots, p)$ factor loadings:

$$\Lambda_l = \rho_l W \Lambda_l + \eta_l, \tag{8}$$

where $\Lambda_l = (\lambda_{1,l}, \ldots, \lambda_{n,l})'$ is an n-dimensional vector of loadings, $R = \text{diag}(\rho_1, \ldots, \rho_p)$ measures the strength of spatial dependence for each vector of loadings $l = (1, \ldots, p)$, W is a row-normalised contiguity weight matrix encoding the connectivity structure between counties, and $\eta_l \sim N(0, \sigma^2_{\eta,l} I_n)$ represents the spatial innovations affecting the sensitivity of consumption growth to macroeconomic factors.

The temporal evolution of each dimension $F_{t,l}$ $(l = 1, \ldots, p)$ of the factor F_t is governed by a first-order autoregressive process:

$$F_{t,l} = \phi_l F_{t-1,l} + \xi_{t,l}, \tag{9}$$

where $\Phi = \text{diag}(\phi_1, \ldots, \phi_p)$ measures the strength of time dependence and $\xi_t \sim N(0, \Sigma_\xi)$ with

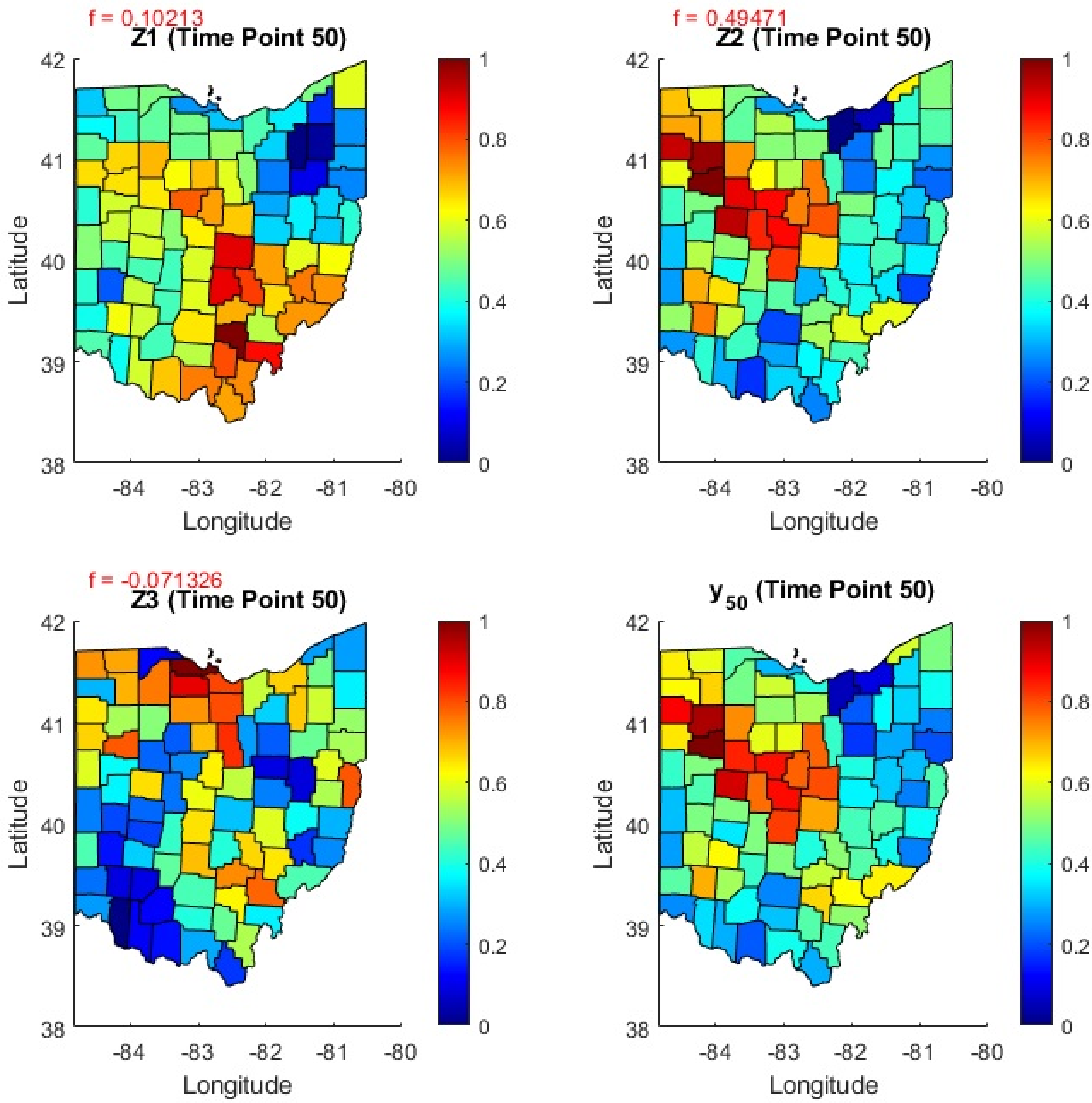
$\Sigma_\xi = \mathrm{diag}(\sigma^2_{\xi,1}, \ldots, \sigma^2_{\xi,p})$ represents economic shocks over time; **the diagonal restrictions on Φ and Σ_ξ are adopted for tractability as discussed in Section 3.3.**

Two modelling decisions require justification. First, the spatial autoregressive structure is introduced in the factor loadings rather than in the idiosyncratic errors. Counties are affected by the same aggregate shocks but differ in how strongly they respond, and these response patterns are spatially correlated across neighbours. Second, the row-normalised contiguity weight matrix W encodes the assumption that geographic proximity proxies for economic integration, standard in the regional economics literature (LeSage and Pace, 2009).

The following illustrative example makes these ideas concrete. We present a brief illustrative example for the $n = 88$ counties of Ohio with $T = 50$ time periods and $p = 3$ latent factors. The simulation parameters are $\rho_l = \phi_l = 0.8$ for all $l = 1, 2, 3$ (homogeneous across factors for this illustration; the model imposes no such restriction in general), $\Sigma_\xi = I_3$, and $\sigma^2_i = 0.1$ for all i. Factors are drawn from Equation (7) with $\Phi = \mathrm{diag}(0.8, 0.8, 0.8)$, so each factor evolves as an independent AR(1) with no cross-factor dynamics. Spatial loadings Λ_l are drawn from the prior $\lambda_l \sim N(0, \sigma^2_{\eta,l}(B_l' B_l)^{-1})$ where $B_l = I_n - \rho_l W$, and idiosyncratic errors from $\epsilon_{it} \sim N(0, 0.1)$. The simulated observed variable is generated as: $y_{it} = \sum_{l=1}^{3} \lambda_{il} F_{t,l} + \epsilon_{it}$. This corresponds to the latent factor component of Equation (6).

Each map in Figure 1 displays $Z_{t,l} = \lambda_l F_{t,l}$, the contribution of each factor to the observed outcome at $t = 50$. The final panel shows $\hat{y}_{50} = \sum_{l=1}^{3} \hat{F}_{50,l} \cdot \hat{\Lambda}_l$, the posterior mean reconstruction mapped across the 88 counties. The first factor's impact is concentrated in the central and southeastern regions, the second dominates the central and northeastern areas, and the third exhibits peak values in the northern counties. The spatial pattern of $\hat{y}_{50}$ closely mirrors that of the second factor, reflecting its dominant role ($F_{50,2} = 0.49$).

Figure 1: Spatial dynamic factor model simulation for $n = 88$ Ohio counties at $t = 50$. Each panel shows $Z_{t,l} = \hat{F}_{50,l} \cdot \hat{\Lambda}_l$ for $l = 1, 2, 3$ and the posterior mean reconstruction $\hat{y}_{50}$. Parameters: $\rho_l = \phi_l = 0.8$, $\Sigma_\xi = I_3$, $\sigma_i^2 = 0.1$.



The simulation illustrates the key feature of the model: geographic spillovers are identified through heterogeneity in shock sensitivity across counties, captured

by the spatially correlated loadings Λ_i, rather than through residual spatial correlation in the errors. The estimation procedure that exploits the banded precision matrix arising from the first-order Markovian factor dynamics is described in Section 4.

4 Estimation procedure

To estimate the spatial dynamic factor model developed in Section 3, we cast the empirical specification as a dynamic factor framework. For $t = 1, \ldots, T$ and $i = 1, \ldots, n$, the observation equation is:

$$y_t = \mu + \delta y_{t-1} + X_t \gamma + \Lambda F_t + \epsilon_t, \quad \epsilon_t \sim N(0, \Sigma), \tag{10}$$

where y_t denotes the vector of county-level spending index values at time t, constructed following Chetty et al. (2024) as detailed in Section 6 and the online supplement Section B. The index measures the deviation of credit and debit card-based spending from its pre-pandemic seasonal baseline; the $\Delta \ln c_t$ notation of Section 3 therefore serves as a conceptual approximation, and all estimation is conducted on y_t directly. The lagged term $\delta y_{i,t-1}$ captures consumption persistence from habit formation and adjustment costs (Carroll, 2000; Fuhrer, 2000), and also accounts for the mechanical persistence in the index construction. The matrix $X_t = (X_{1,t}, \ldots, X_{n,t})'$ contains time-varying explanatory variables including pandemic indicators, policy measures, and economic metrics (described in Section 6). Income growth, which enters the theoretical framework through the rule-of-thumb term $\nu \Delta \ln y_{it}$ in Equation (6), is proxied through per capita income and percentage income change variables in X_t. The idiosyncratic error covariance is $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$.

A major challenge in factor analysis is ensuring the model's uniqueness and avoiding identification issues. In the proposed model, the product ΛF_t is uniquely identified, but the individual components Λ and F_t are not, as the relationship $\Lambda F_t = (\Lambda K^{-1})(K F_t)$ holds for any invertible matrix K. The scale ambiguity is most naturally resolved by imposing the classical lower triangular normalisation, fixing the first p rows of Λ to the identity matrix I_p, and we impose constraints on Λ as discussed in the online supplement Section D.1.

Stacking observations over time and applying properties of the Kronecker product, we use the identity:

$$\operatorname{vec}\begin{pmatrix} \Lambda F_1 \\ \Lambda F_2 \\ \vdots \\ \Lambda F_T \end{pmatrix} = (F \otimes I_n)\operatorname{vec}(\Lambda), \tag{11}$$

where F $(T \times p)$ denotes the matrix with F_t' as its t-th row. We distinguish three related objects throughout this section: F_t $(p \times 1)$ denotes the factor vector at a single time point t; F $(T \times p)$ denotes the matrix with F_t' as its t-th row, used in the Kronecker identity above; and $\mathbf{f} = \operatorname{vec}(\mathbf{F}')$ $(Tp \times 1)$ denotes the column-stacked vector, where each column of $\mathbf{F}'$ contains the time series of one factor, appearing in the precision matrix and simulation smoother of Section 4.3. Here Λ is of dimension $n \times p$, with p denoting the number of factors.

Using this identity, the stacked observation equation can be written as:

$$y = X\beta + (F \otimes I_n)\operatorname{vec}(\Lambda) + \epsilon, \quad \epsilon \sim N(0, I_T \otimes \Sigma), \tag{12}$$

where $y = (y_1', \ldots, y_T')'$ is of dimension $nT \times 1$, and $X = (x_1', \ldots, x_T')'$ is of dimension $nT \times (k+2)$, with each $x_t = [\iota_n, y_{t-1}, X_t]$ and ι_n being an $n \times 1$ vector of ones. The coefficient vector β has dimension $(k+2) \times 1$.

4.1 Likelihood function

Given the stacked observation equation (12) and the assumption that $\epsilon \sim N(0, I_T \otimes \Sigma)$, the idiosyncratic errors ϵ_t are independent across time periods. **The joint likelihood therefore factors as $\prod_{t=1}^{T} p(y_t|F_t, \beta, \Sigma)$, where each term is multivariate normal with mean** $\mu + \delta y_{t-1} + X_t\gamma + \Lambda F_t$ and covariance Σ. Multiplying across T periods and collecting the determinant and quadratic form terms, noting that $|I_T \otimes \Sigma| = |\Sigma|^T$ and that the block-diagonal structure of $I_T \otimes \Sigma^{-1}$ allows the exponent to be written in terms of the stacked residual, yields the joint likelihood:

$$\begin{aligned} p(y|\Theta, F, \Lambda, X) \;&=\; (2\pi)^{-nT/2}|\Sigma|^{-T/2} \\ &\quad \times \exp\left\{-\frac{1}{2}(y - X\beta - (F \otimes I_n)\mathrm{vec}(\Lambda))'(I_T \otimes \Sigma^{-1})(y - X\beta - (F \otimes I_n)\mathrm{vec}(\Lambda))\right\}, \end{aligned} \tag{13}$$

Here vec(Λ) is the $np \times 1$ column vector obtained by stacking the columns of the $n \times p$ matrix Λ, and $\Theta = (\beta, R, \Phi, \Sigma, \Sigma_\xi, \Sigma_\eta)$.

4.2 Prior Distributions

To simplify the estimation process, conditionally conjugate prior distributions are assigned to all parameters defining the dynamic factor model. In contrast, for the parameters governing the spatial processes, a draw-by-inversion method based on univariate integration, as described in LeSage and Pace (2009), is used.

The prior for the common dynamic factors is given by the time evolution equation and is completed by assuming that $F_1 \sim N(m_1, C_1)$, where m_1 and C_1 are known hyperparameters. Independent prior distributions are assigned to the variance parameters of the idiosyncratic errors, factor disturbances, and spatial processes. Specifically, for $i = 1, \ldots, n$, the variance of the idiosyncratic errors follows an inverse gamma distribution, $\sigma_i^2 \sim IG(n_{\epsilon,i}/2, s_{\epsilon,i}/2)$. Similarly, for $l = 1, \ldots, p$, each variance of the factor disturbances is modeled as $\sigma_{\xi,l}^2 \sim$

$IG(n_{\xi,l}/2, s_{\xi,l}/2)$, and the variances of the spatial processes follow $\sigma^2_{\eta,l} \sim IG(n_{\eta,l}/2, s_{\eta,l}/2)$. The terms $n_{\epsilon,i}, n_{\xi,l}, n_{\eta,l}, s_{\epsilon,i}, s_{\xi,l}, s_{\eta,l}$ are known hyperparameters.

The time dynamic process is assumed to be stationary hence $\phi_l \sim N_{tr_{(-1,1)}}(m_\phi, S_\phi)$, where $N_{tr_{(a,b)}}(.)$ refers to the Normal distribution truncated to the interval $[a, b]$.

The prior distribution for each ρ_l is defined over the interval $(1/\psi_{\min}, 1)$, where $\psi_{\min}$ denotes the minimum eigenvalue of the row-normalized spatial weight matrix W. A normal prior is assigned to β, such that $\beta \sim N(\beta_0, B)$. To ensure a diffuse prior, the mean is set to $\beta_0 = 0$, and large values are chosen for the diagonal elements of the variance matrix B. The use of inverse gamma distributions for variance parameters allows for a straightforward posterior update, thereby facilitating computational efficiency in the estimation procedure.

4.3 Efficient Estimation of Latent States

Following Fahrmeir and Kaufmann (1991), the dynamic process for the factors, as described in Equation (9), can be rewritten as $H\mathbf{f} = \xi$, where H is a $Tp \times Tp$ block matrix defined as:

$$H_{i,j} = \begin{cases} I_p & \text{if } i = j, \\ -\Phi & \text{if } i = j + 1, \qquad i, j = 1, 2, \ldots, T, \\ 0 & \text{otherwise,} \end{cases}$$

with I_p denoting the $p \times p$ identity matrix. The vector of disturbances, $\xi = (\xi_1', \ldots, \xi_T')'$, follows a normal distribution with $\xi_t \sim N(0, \Sigma_\xi)$ for $t = 2, \ldots, T$, where $\Sigma_\xi = \text{diag}(\sigma^2_{\xi,1}, \ldots, \sigma^2_{\xi,p})$. Using a Prais-Winsten type transformation, the initial disturbance is distributed as $\xi_1 \sim N(0, \Sigma_{\xi,1})$, where

$$\Sigma_{\xi,1} = \text{diag}\left(\frac{\sigma^2_{\xi,1}}{1 - \phi_1^2}, \ldots, \frac{\sigma^2_{\xi,p}}{1 - \phi_p^2}\right).$$

Thus, the $Tp \times Tp$ precision matrix of $\mathbf{f}$ is given by:

$$M = H' S^{-1} H, \tag{14}$$

where $S = \mathbf{diag}(\Sigma_{\xi,1}, \Sigma_{\xi}, \ldots, \Sigma_{\xi})$. Because Φ is diagonal (see Section 3.3), H is block-bidiagonal and M inherits a block-banded structure with bandwidth p (Chan and Jeliazkov, 2009; Beyeler and Kaufmann, 2021); the full sparsity implications and implementation details are discussed in Section 4.3.1.

Given the Gaussian likelihood of the observed data and the prior precision structure of the latent factors, the conditional posterior distribution of the factor vector $\mathbf{f}$, conditional on the data y, the factor loadings Λ, and the model parameters Θ, is Gaussian. This follows directly from Bayes' theorem, under which the posterior distribution $p(\mathbf{f} \mid y, \Lambda, \Theta)$ is fully characterized by a precision matrix Ω and a posterior mean $\hat{\mathbf{f}}$, given by

$$\Omega = M + I_T \otimes (\Lambda' \Sigma^{-1} \Lambda)$$

$$\hat{\mathbf{f}} = \Omega^{-1} (I_T \otimes \Lambda' \Sigma^{-1})(y - X\beta).$$

Since $I_T \otimes (\Lambda' \Sigma^{-1} \Lambda)$ is block-diagonal with $p \times p$ blocks, and M is block-banded with bandwidth p, Ω inherits this block-banded structure with bandwidth p, enabling the aforementioned computational advantages. Notably, our method accommodates both proper and improper prior distributions. In the case of a diffuse prior (improper), M becomes rank-deficient, which is easily addressed by adjusting the dimensions of H and S, thereby ensuring the method's versatility. This section extends McCausland et al. (2009) to express an efficient simulation method for the density $\pi(\mathbf{f}|y, \Theta, \Lambda)$, crucial for estimating latent states in models with banded precision matrices.

4.3.1 Modeling dynamic process using Cholesky decomposition

To leverage the banded structure of the precision matrix Ω, we avoid direct inversion and instead use its Cholesky decomposition. Let C be the upper triangular factor such that $C'C = \Omega$. The posterior mean $\hat{\mathbf{f}}$ is then obtained by solving the linear system:

$$C'C\hat{\mathbf{f}} = (I_T \otimes \Lambda'\Sigma^{-1})(y - X\beta). \tag{15}$$

This solution is efficiently computed using forward and backward substitution. Similarly, to draw a random sample from the distribution $N(\hat{\mathbf{f}}, \Omega^{-1})$, we generate $u \sim N(0, I)$, solve $Cx = u$ for x via back-substitution, and obtain $\mathbf{f} = \hat{\mathbf{f}} + x$.

The precision matrix Ω is block-banded with bandwidth p and is stored and operated on as a sparse matrix throughout. In our MATLAB implementation, standard sparse matrix routines (`speye`, `sparse`, `chol`) are used to exploit this structure automatically. Because Ω is block-banded, its exact Cholesky factor C is also block-upper-triangular with the same bandwidth p, a standard result for banded positive definite matrices, so fill-in remains confined within the original bandwidth. The triangular factor is therefore sparse with the same pattern as Ω, and the forward and backward substitutions for $\hat{\mathbf{f}}$ and the simulation draw each cost $O(Tp^2)$ operations. The banded Cholesky factorization itself costs $O(Tp^3)$ rather than $O(T^3p^3)$ for a dense Cholesky. An incomplete Cholesky is therefore unnecessary, since the exact factorization already preserves sparsity at no additional cost. A figure illustrating the sparsity patterns of Ω and C is provided in the online supplement Section D.4.2.

4.3.2 Spatial loadings

As defined in Equation (8), the lth column of Λ follows a spatial autoregressive process such that $\Lambda_l \sim N(0, \sigma^2_{\eta,l} S_l S_l')$, where $S_l = (I_n - \rho_l W)^{-1}$. In matrix form, the factor loadings satisfy $\text{vec}(\Lambda) = (I_{np} - R \otimes W)^{-1}\text{vec}(\eta)$, where $\eta \sim N(0, \Sigma_\eta \otimes I_n)$ with $\Sigma_\eta = \text{diag}(\sigma^2_{\eta,1}, \ldots, \sigma^2_{\eta,p})$.

By introducing spatial dependence into the loadings, separable covariance functions for spatiotemporal processes can be expressed as either the product or sum of a purely spatial covariance function and a purely temporal covariance function. Specifically, let y_{it} be a random process indexed by both space and time. Under the proposed model, when $p = 1$ and $\beta = 0$, the covariance structure simplifies to $Cov(y_{it}, y_{j,t+h}) = \sigma_\xi^2 \phi^h (1-\phi^2)^{-1}(\sigma_\eta^2 (SS')_{ij})$ which confirms that both the spatial and temporal covariance functions are separately identifiable. More generally, for $p > 1$, the covariance extends to $Cov(y_{it}, y_{j,t+h}) = \sum_{l=1}^p \sigma_{\xi,l}^2 \phi_l^h (1-\phi_l^2)^{-1}(\sigma_{\eta,l}^2 (S_l S_l')_{ij})$. This result highlights a key property of the proposed model: whenever the number of common factors exceeds one, the resulting covariance function becomes non-separable.

4.4 Posterior Inference

Posterior inference for the proposed class of spatial dynamic factor models is carried out using a Markov Chain Monte Carlo (MCMC) algorithm, assuming a known number of common factors, denoted as p. Given p, the joint posterior distribution of $(F, \beta, \Theta, \Lambda)$ can be expressed as

$$\begin{aligned} p(F, \beta, \Theta, \Lambda | y) \propto & \prod_{t=1}^T p(y_t | F_t, \beta, \sigma) p(F_1 | m_1, C_1) \prod_{t=2}^T p(F_t | F_{t-1}, \Phi, \Sigma_\xi) \\ & \times p(\beta) \prod_{i=1}^n p(\sigma_i^2) \prod_{l=1}^p \left\{ p(\Lambda_l | \rho_l, \sigma_{\eta,l}^2) p(\rho_l) p(\sigma_{\eta,l}^2) p(\phi_l) p(\sigma_{\xi,l}^2) \right\}. \end{aligned}$$

This posterior distribution is analytically intractable and we therefore estimate it via a Gibbs sampler. While the common factors are typically sampled using the forward filtering backward sampling (FFBS) algorithm (Carter and Kohn, 1994), we instead exploit the banded structure of the precision matrix Ω for improved computational efficiency. We draw the regression coefficients β from a normal posterior via the Woodbury identity, and the spatial factor loadings Λ_l ($l = 1, \ldots, p$) from normal full conditionals. The latent factors f are drawn jointly using the banded Cholesky simulation smoother described in Section 4.3.1. The

spatial autoregressive parameters ρ_l are sampled via draw-by-inversion (LeSage, 1997), and the spatial error variances $\sigma^2_{\eta,l}$ from inverse-gamma posteriors. The factor autoregressive parameters ϕ_l require a Metropolis-Hastings step, while the factor innovation variances $\Sigma_\xi = \mathrm{diag}(\sigma^2_{\xi,1}, \ldots, \sigma^2_{\xi,p})$ and idiosyncratic variances σ^2_i are drawn from inverse-gamma posteriors. Full derivations of all conditional distributions are provided in the online supplement Section D.4.

5 Monte Carlo Simulations

This section presents Monte Carlo simulations designed to assess the performance of our spatial dynamic factor model compared to the traditional Bayesian framework in Bernanke et al. (2005) (henceforth BBE), which estimates latent factors using the Kalman filter and smoother within a Gibbs sampling procedure. In particular, we focus on their one-step Bayesian estimation approach, which uses Gibbs sampling to jointly estimate latent factors and VAR dynamics via a state-space model (see in the online supplement Section C for the Bayesian implementation). The objective is to evaluate whether incorporating spatial dependence in factor loadings improves factor recovery, forecasting accuracy, and structural inference, particularly when cross-sectional dependence is present.

5.1 Simulation Design

The simulation design follows the general dynamic factor framework of Equation (10), with factor dynamics defined in Equation (7) and the spatial loading structure of Equation (8). For the simulation, we use a simplified version without the regression component ($\mu = 0$, $\delta = 0$, $X_t = 0$), focusing on the core factor structure with $p = 3$ latent factors. The primary distinction across scenarios

is the presence or absence of spatial dependence in factor loadings. The three equations governing the simulation design are:

$$Y_t = \Lambda F_t + \epsilon_t, \quad \epsilon_t \sim N(0, \Sigma), \tag{16}$$

where the diagonal elements of Σ **are drawn independently from** $\text{Uniform}(0.25, 0.75)$ **to simulate heteroskedasticity;**

$$F_t = \Phi F_{t-1} + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \Sigma_\xi), \tag{17}$$

with $\Phi = \text{diag}(0.8, 0.8, 0.8)$ **and** $\Sigma_\xi = \text{diag}(1, 1, 1)$**; and**

$$\text{vec}(\Lambda) = (I_{np} - R \otimes W)^{-1} \text{vec}(\eta), \tag{18}$$

where $R = 0.8 I_p$ **in the spatial dependence scenario and** $R = 0$ **otherwise,** W **is a spatial weight matrix based on Delaunay triangulation of randomly assigned geographic coordinates, and** $\eta \sim \mathcal{N}(0, \Sigma_\eta \otimes I_n)$ **with** $\Sigma_\eta = I_p$**, so that the spatial innovations are independent across factors with unit variance.** The simulations are conducted for different sample sizes ($n = 5, 50, 500$) with $T = 200$ time periods, discarding the first 50 periods to eliminate initialization effects. **All results are averaged over** $M = 30$ **Monte Carlo replications.** The main evaluation metric is the Mean Squared Error (MSE), defined for a parameter θ as:

$$\text{MSE}(\theta) = \frac{1}{M} \sum_{m=1}^{M} \left(\hat{\theta}^{(m)} - \theta_0 \right)^2,$$

where $\hat{\theta}^{(m)}$ **is the posterior mean from replication** m **and** θ_0 **is the true parameter value used in the data-generating process.**

5.2 Simulation Results

The results are summarized in Table 1. The specifications differ only in the treatment of spatial dependence, which is either imposed ($R = 0.8 I_3$**) or ruled**

out ($R = 0I_3$).

Table 1: MSE Comparison of Kalman Filter and Gibbs Sampler (Spatial vs. Non-Spatial)

Sample Size	True Process	With Spatial Dependence ($R = 0.8I_3$)		No Spatial Dependence ($R = 0I_3$)	
		Gibbs	Kalman	Gibbs	Kalman
$n = 5$	Λ	0.015	0.019	0.005	0.008
	f	0.076	0.114	0.087	0.136
	Σ_ξ	0.003	0.006	0.003	0.007
	Σ	0.002	0.048	0.002	0.012
	R	0.519	—	—	—
	Φ	0.003	0.003	0.003	0.004
$n = 50$	Λ	0.019	0.028	0.009	0.013
	f	0.007	0.047	0.009	0.043
	Σ_ξ	0.001	0.002	0.001	0.002
	Σ	0.001	0.006	0.001	0.002
	R	0.018	—	—	—
	Φ	0.002	0.003	0.003	0.006
$n = 500$	Λ	0.019	0.024	0.008	0.010
	f	0.005	0.041	0.004	0.030
	Σ_ξ	0.001	0.012	0.001	0.001
	Σ	0.001	0.003	0.001	0.002
	R	0.002	—	—	—
	Φ	0.003	0.004	0.001	0.002

Table 1 demonstrates that the Bayesian spatial model systematically outperforms the Kalman filter when spatial dependence is present. The MSE for factor loadings, latent factors, and variance components is substantially lower under the Gibbs sampler in the spatial setting. When spatial dependence is absent, both methods perform comparably, though the Gibbs sampler maintains a slight advantage in high-dimensional settings ($n = 500$).

For the Kalman filter comparison, we implement the BBE (Bernanke et al., 2005) framework using the standard forward Kalman filter and Rauch-Tung-Striebel smoother (detailed in the online supplement Section C), with the same Gibbs sampler structure for all parameters except the factors. This is initialized at principal-component estimates of the loadings and factors. Both methods are run for the same number of MCMC iterations (5,000 draws, 1,000 burn-in) to ensure a fair computational comparison. The banded Cholesky simulation smoother costs $\mathcal{O}(Tp^3)$ for the factorization and $\mathcal{O}(Tp^2)$ for the forward and backward substitutions per iteration, compared to $\mathcal{O}(Tn^3)$ for the Kalman-based approach, where the dominant cost is the $\mathcal{O}(n^3)$ inversion of the $n \times n$ innovation covariance $S_t = \Lambda P_{t|t-1}\Lambda' + \Sigma$ at each of the T filter steps. This advantage grows with n, making the proposed approach particularly well-suited to large cross-sectional settings such as the $1{,}679$-county application in Section 7.

6 Data Description

This study employs a granular county-level dataset to examine the impact of the COVID-19 pandemic, government interventions, and fiscal stimulus on consumer spending behavior. The dataset spans from January 2019 through December 2022, integrating weekly economic indicators, public health metrics, and policy measures across 1,679 U.S. counties. The estimation sample consists of 154 weekly observations beginning in the third week of January 2020 and ending in December 2022, with January 2020 used as the baseline for index construction. The inclusion of $y_{i,t-1}$ as a regressor reduces the effective estimation sample to $T = 153$ time periods. In addition, to examine regional disparities in baseline

consumption and policy responses, counties are classified as urban or rural using designations from the Census Bureau (*Rural*).

The primary dependent variable, *Consumption*, is the Opportunity Insights Economic Tracker spending index constructed following Chetty et al. (2024), based on anonymized credit and debit card spending data from Affinity Solutions, which captures nearly 10% of debit and credit card spending in the U.S. The index measures the deviation of spending from its pre-pandemic seasonal path: each week's spending in 2020 onwards is expressed relative to the corresponding week in 2019 to remove seasonal variation, then indexed to January 2020 as the baseline. Positive values indicate spending above the pre-pandemic seasonal path; negative values indicate spending below it. A full derivation of the index in log terms is provided in the online supplement Section B.

The control variables X_{it} fall into four groups. Labor market conditions are captured by initial unemployment claims (*initclaims*) and employment growth (*emp*). Income variables include per capita income (*perCapita*) and income percentage changes (*percentChange*), which proxy for the income growth term in the theoretical framework of Section 3. Financial conditions are measured by the S&P 500 index return (*SP500*), federal interest rates (*FEDinterest*), consumer sentiment toward borrowing conditions (*Sentimentinterest*), and inflation expectations through the Consumer Price Index (*CPI*). **Policy variables include Economic Impact Payments (EIP1, EIP2, and EIP3), measuring the total amount disbursed at the state level for each payment round under the CARES Act, sourced from IRS distribution records; these variables interact with *Rural* to allow differential stimulus responses by county type.**

To capture the direct effects of the pandemic on consumer confidence and mobility, several COVID-19-related indicators are included: death rates (*New_death*), and full vaccine coverage (*New_Fullvaccine*). Government containment measures are captured by stay-at-home orders (*stayhome*) and business closure indicators (*Busclosure*).

Table 2 presents a summary of all variables. Figure 2 depicts differences in average consumption trends between urban and rural counties. Two observations support the claim that pre-COVID consumption patterns were similar across regions. First, because the index is constructed using January of each year as a within-year baseline and expressed as a year-over-year ratio, all series start near zero in January 2020 by construction and seasonal differences are removed by design. Second, and more directly visible in the figure itself, the first approximately ten weeks of the estimation sample (January to mid-March 2020, prior to the pandemic shock) show virtually identical consumption trajectories across urban and rural counties, providing direct visual evidence of pre-shock comparability without requiring pre-2020 data. With the emergence of the pandemic, urban areas then experienced a substantially steeper decline in consumption.

To handle missing data, we include dummy variables for each covariate with missing values, denoted by the "_NA" suffix (e.g., *emp_NA*, *initclaims_NA*). These indicators capture the presence of missing observations, helping mitigate biases due to non-random missingness in the data.

7 Empirical Framework

Using the county-level dataset described in Section 6, this study employs a dynamic factor model to analyze consumption spending during the COVID-19 pandemic. The model, specified for weekly observations from January 2020 to December 2022, is defined as:

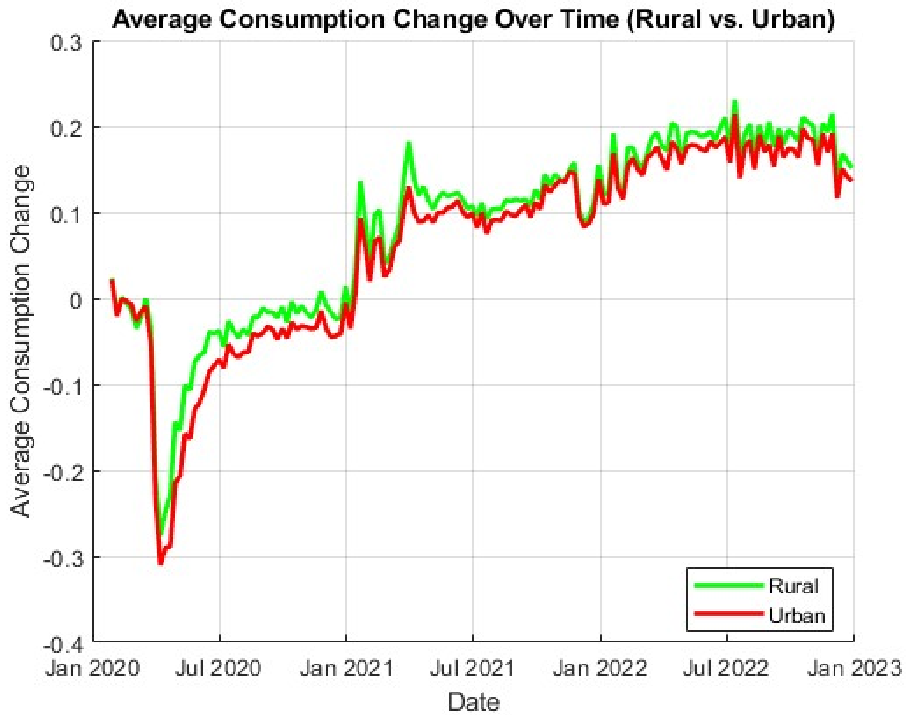
$$y_{it} = \mu + \delta y_{i,t-1} + X_{it}\gamma + \Lambda_i' F_t + \epsilon_{it}, \tag{19}$$

Table 2: Summary Statistics for Empirical Variables

Variable	Description	Unit	Mean	Std. Dev.	Min	Max	Source
Consumption	Change in spending	% change from Jan. 2020	0.0710	0.1786	-0.9329	0.6320	Economic Tracker
New_death	New COVID-19 deaths	Deaths per 100K	0.3475	0.6693	0	17.8	Economic Tracker
New_Fullvaccine	New vaccinations	Vaccinations per 100K	0.0503	0.1304	0	8.245	Economic Tracker
stayhome	Stay-at-home policy indicator	Binary (0/1)	0.0365	0.1875	0	1	Economic Tracker
Busclosure	Business-closure policy indicator	Binary (0/1)	0.0302	0.1713	0	1	Economic Tracker
initclaims	Initial unemployment claims	% of population	0.1690	0.5446	0	30.5	Economic Tracker
Rural	Rural county indicator	Binary (0/1)	0.4264	0.4946	0	1	U.S. Census Bureau
perCapita	Per capita income	Annual USD	53563.000	18712.000	0	406050	U.S. Census Bureau
percentChange	Annual percentage change in income	%	5.5198	4.3122	-23	47.2	U.S. Census Bureau
emp	Employment growth	% change from Jan. 2020	-0.0128	0.1542	-1	2.33	BLS
SP500	S&P 500 return	%	0.1798	3.3903	-14.537	12.341	Financial data
FEDinterest	Federal funds rate	%	0.7150	1.1609	0.05	4.1	Federal Reserve
Sentimentinterest	Consumer sentiment on interest rates	%	0.2960	0.0341	0.22	0.39	FRBNY SCE
CPI	Consumer Price Index	Index	0.3574	0.1733	0.0906	0.6471	BLS
EIP1	Economic Impact Payment 1	USD, scaled by 100 million	0.0757	0.0712	0	0.2960	IRS administrative data
EIP2	Economic Impact Payment 2	USD, scaled by 100 million	0.0289	0.0366	0	0.1541	IRS administrative data
EIP3	Economic Impact Payment 3	USD, scaled by 100 million	0.0752	0.1040	0	0.4477	IRS administrative data

which is the empirical counterpart of Equations (6)-(9), with factor loadings Λ_i following the spatial process of Equation (8). Here y_{it} is the Opportunity Insights spending index for county i at time t (described in Section 6 and supplement Section B), measuring the deviation of card-based spending from its pre-pandemic seasonal baseline. Each county $i = (1, \ldots, n = 1679)$ contributes a weekly observation of this index. The spatial weight matrix W is constructed using the 8 nearest neighbours for each county, based on county centroid distances and row-normalised so that each row sums to one. A nearest-neighbour specification is adopted rather than contiguity because the sample covers only 1,679 of the 3,143 U.S. counties, meaning that contiguous counties may not always be present in the dataset. The vector X_{it} comprises the explanatory variables described in Table 2, including pandemic-related indicators, policy variables, and economic metrics.

Figure 2: Changes in Average Consumption: Rural vs. Urban Counties



The term $\Lambda_i' F_t$ captures unobserved spatial and temporal dependencies. The loadings $\Lambda = (\Lambda_1, \ldots, \Lambda_n)'$, modeling unobserved spatial dependence, are defined in Equation (8), and the time dynamic factors F_t, representing common temporal shocks, are defined in Equation (9). This framework allows for the examination of both direct and indirect effects of the pandemic and policy responses on consumer spending, while controlling for unobserved factors through the dynamic factor structure.

7.1 Estimation results

The results of the MCMC sampling for the spatial regression model with latent factors and spatial dependence are summarized below. The MCMC sampling ran for 100,000 iterations, with 10,000 iterations discarded as burn-in, and successfully converged for all parameters. **Throughout, we define a coefficient as statistically significant if its 95% HPD**

interval excludes zero. The results are as below:

Table 3: Estimation Results (Number of Iterations = 100,000)

Variable	Estimate	Std. Deviation	Lower Bound	Upper Bound
constant	-0.131	0.004	-0.138	-0.123
y_{t-1}	0.492	0.003	0.486	0.497
New_death	-0.008	0.002	-0.012	-0.003
New_death_NA	0.004	0.001	0.002	0.007
New_Fullvaccine	0.003	0.002	-0.001	0.007
New_Fullvaccine_NA	0.002	0.002	-0.001	0.005
Busclosure	-0.017	0.002	-0.021	-0.014
Busclose_NA	-0.001	0.001	-0.003	0.002
initclaims	-0.002	4×10^{-4}	-0.003	-0.001
initclaims_NA	0.002	0.001	2×10^{-4}	0.004
emp	0.014	0.003	0.008	0.019
emp_NA	0.003	0.001	0.001	0.004
EIP1	-0.103	0.015	-0.132	-0.074
EIP2	-0.094	0.026	-0.144	-0.043
EIP3	-0.009	0.008	-0.026	0.007
Sentimentinterest_2020	-0.119	0.012	-0.144	-0.095
Sentimentinterest_2021	0.075	0.013	0.051	0.100
Sentimentinterest_2022	0.124	0.011	0.101	0.145
SP500_2020	-0.001	0.001	-0.002	1×10^{-4}
SP500_2021	2×10^{-4}	0.001	-0.001	0.001
SP500_2022	-0.002	4×10^{-4}	-0.002	-0.001
Percentchange_income_2020	2.145	1.308	-0.437	4.689
Percentchange_income_2021	-0.805	1.501	-3.742	2.132
Percentchange_income_2022	5.127	1.359	2.460	7.788
EIP1_rural	0.146	0.019	0.108	0.185
EIP2_rural	-0.090	0.035	-0.157	-0.018

Table 3: (continued)

Variable	Estimate	Std. Deviation	Lower Bound	Upper Bound
EIP3_rural	-0.008	0.012	-0.031	0.015
ϕ	0.958	0.024	0.913	0.999
σ_ξ^2	0.008	0.001	0.006	0.010
σ_η^2	0.264	0.035	0.199	0.335
ρ	0.458	0.047	0.367	0.549

Note: Lower and Upper Bounds are for the 95% High Posterior Density Interval (HPD Interval)

Note that the empirical illustration relies on one-dimensional loadings ($p = 1$), as additional dimensions did not yield statistically meaningful loadings and did not produce a substantial increase in the log marginal likelihood.

As a robustness check, we estimated a non-spatial version of the model (imposing $\rho = 0$) and compared the resulting factor loadings across all 1,679 counties. While posterior means are broadly stable across specifications, visible deviations arise for counties with stronger geographic spillovers, reflecting the additional information captured by the spatial autoregressive structure. The spatial model also yields tighter posterior distributions, with the average posterior standard deviation decreasing by approximately 6.6% (from 0.066 to 0.062). Full details and a graphical comparison are provided in the online supplement Section F.1.

7.2 Empirical findings

We analyze credit card spending data to uncover the drivers of consumption patterns during COVID-19, focusing on health outcomes, income changes, stimulus effects, and spatiotemporal dynamics. New COVID-19-related deaths are significantly negatively associated with spending (–0.008), suggesting heightened uncertainty and voluntary mobility reduction curtailed household expenditure, consistent with Goolsbee and Syverson (2021), who show consumer behavior, not lockdowns, drove most early declines. Their mobility data reveal that only 7 percentage points of reduced visits were policy-induced, the rest due to voluntary risk avoidance. Our findings echo this: while business closures show a negative spending effect, health concerns were more influential.

Higher unemployment claims also reduce spending (–0.002), underscoring the impact of labor market stress. Consumer sentiment about interest rates influenced behavior: in 2020, pessimism suppressed spending (–0.119), while in 2021 optimism returned (0.075). The relationship between S&P 500 performance and spending is mixed, implying indirect or uneven financial-market effects.

Income effects were insignificant in 2020–21 but significant in 2022 (5.13), suggesting health risks and uncertainty dominated early pandemic behavior, consistent with Chetty et al. (2024). Stimulus payments, particularly the first-round EIP1 (-0.103), had muted effects. This supports survey evidence (Coibion et al., 2025; Parker et al., 2022) that households used early EIPs for savings or debt repayment. However, EIP1 significantly boosted rural spending (0.146), consistent with greater liquidity constraints. This effect faded in later rounds (EIP2 rural: -0.090; EIP3: ≈ 0), consistent with declining marginal propensities to consume (Cox et al., 2020).

Building on Chetty et al. (2024), who show that early declines in U.S. consumer spending during the COVID-19 pandemic were driven more by health concerns than income loss, we use the same high-frequency Opportunity Insights dataset on credit and debit card transactions to explore geographic variation in spending behavior at the county level. While their analysis focuses on income-based heterogeneity, we apply a spatial dynamic factor model to estimate county-specific loadings that reveal substantial regional differences in responsiveness to national economic shocks. These loadings capture unobserved local influences, such as regional policy responses, institutional capacity, or behavioral norms, that shaped consumption patterns beyond what is explained by observable variables like income, pandemic severity, or stimulus exposure.

The temporal evolution of the estimated latent factor, shown in Figure 3, reveals a sharp initial decline in spending, followed by a surprisingly rapid rebound. Although the factor is highly persistent (with an autoregressive coefficient of $\phi = 0.958$), the swift recovery likely reflects the interplay between enduring structural shifts and shorter-term forces, including rising consumer confidence following vaccine distribution and the financial support provided by EIPs. The broader stability of the system and its implications are discussed in the online supplement Section E.

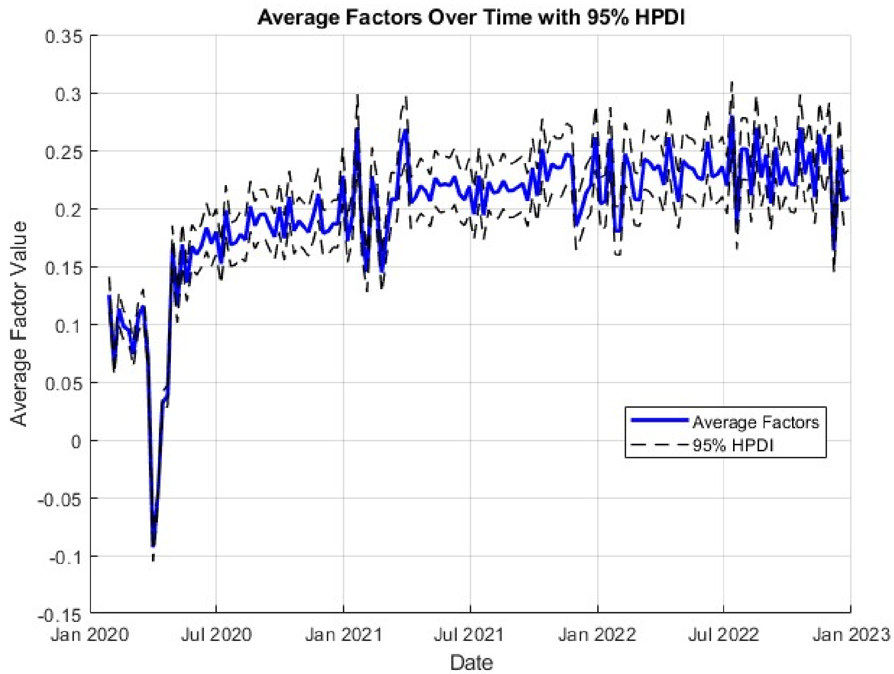
Figure 4 illustrates the spatial distribution of factor loadings, highlighting substantial regional variation in unexplained consumption changes during the COVID-19 pandemic. The estimated spatial autoregressive parameter ($\rho = 0.458$) indicates a strong positive correlation in latent consumption influences among neighboring counties, suggesting that economic structures, policy responses, and pandemic severity contributed to similar consumption patterns across geographically proximate areas.

While rural areas exhibited largely predictable consumption shifts, substantial unexplained variation is concentrated in suburban metropolitan areas, particularly in the Midwest and South. **Michigan and the Indianapolis metropolitan area stand out with a high number of counties showing large deviations from predicted consumption trends. Interestingly, these areas are consistent with regions that experienced significant pandemic-related demographic shifts, including outflows from major urban centres such as Chicago and New York City, though the model cannot attribute causality to these factors directly.** Section F in the online supplement further explores factor loadings across counties and states, linking them to observable and latent drivers. Local policy quality, consumer norms, and mobility appear to shape spending beyond observable variables.

Regions with the most negative loadings, indicating greater-than-expected declines in consumption, are concentrated in Indiana and Ohio, particularly in suburban areas. **This is consistent with local economic disruptions, shifts in employment, or unaccounted regional factors being associated with sharper-than-anticipated spending contractions.** In particular, the Indianapolis metropolitan area exhibits several counties with significant negative loadings, **consistent with the vulnerability of suburban communities to unexpected consumption downturns.** Possible explanations include employment losses in key sectors, disruptions in local service economies, and shifting household financial priorities.

To further understand the sources of regional variation in macroeconomic sensitivity, the online supplement Section F presents a detailed analysis of estimated factor loadings at both the county and state levels. This analysis examines how the loadings relate to observable characteristics (including baseline consumption, exposure to federal stimulus, public health outcomes, and labor market dynamics), while also highlighting the role of unobserved regional factors. The results show that while policy supports and economic conditions help explain much of the variation, latent local characteristics, such as governance quality, consumer behavior, and mobility trends, also play a critical role in shaping consumption responses. This richer interpretation of the factor loadings complements the spatial patterns displayed in Figure 4, providing deeper insight into the drivers of regional divergence in aggregate consumption dynamics.

Figure 3: Factor Estimates with 95% Highest Posterior Density Interval

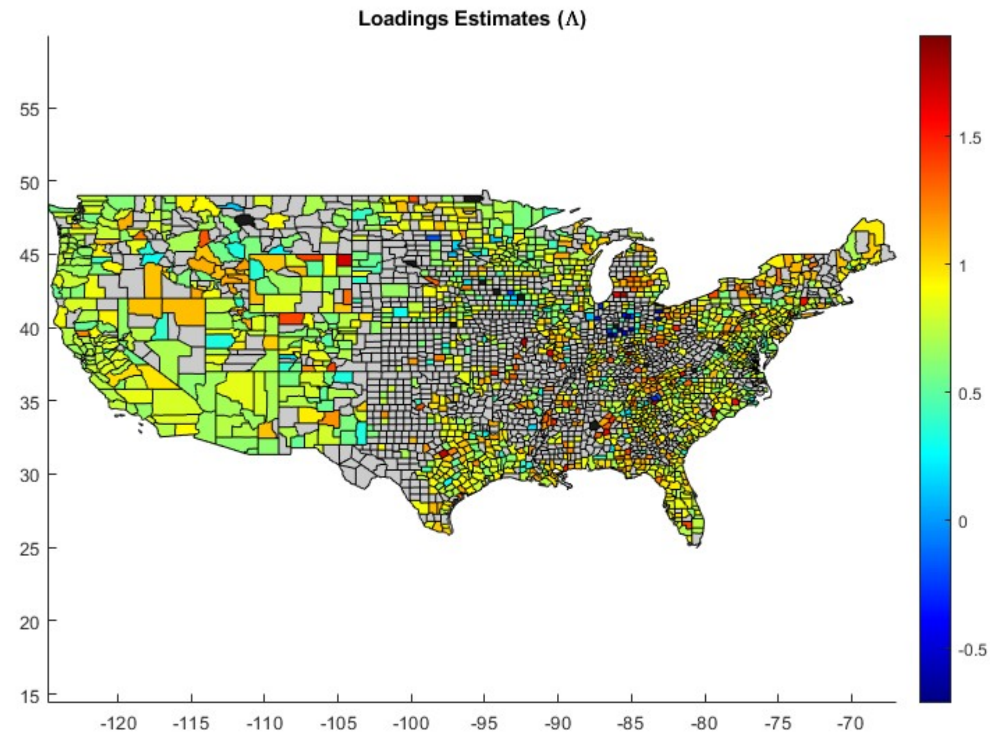


8 Conclusion

This paper examines how the COVID-19 pandemic shaped regional consumption dynamics across the United States. By combining a latent dynamic factor structure with spatial autoregressive dependencies, we estimate county-specific sensitivities to national shocks while explicitly accounting for local spillovers and spatial heterogeneity.

Methodologically, we contribute to the literature on high-dimensional spatiotemporal modeling by introducing a computationally efficient estimation procedure for spatial dynamic factor models. By exploiting the banded structure of the inverse covariance matrix, we achieve scalable inference even in settings with fine spatial resolution and long time series. This facilitates the application of rich spatial models to granular economic data and lays the groundwork for further extensions involving multiple latent factors, time-varying loadings, or dynamic treatment effects.

Figure 4: Spatial Distribution of Loadings (Λ) with Non-Significant Estimates in Black (95% HPD interval includes zero)



Empirically, our results uncover substantial regional heterogeneity in consumption responses that is not captured by observable factors alone. The estimated factor loadings reveal persistent spatial clustering and local variation in exposure to national shocks, with some counties, especially in the Midwest, exhibiting unexpectedly large and sustained declines in consumption. These patterns point to the importance of unobserved institutional, behavioral, or structural features shaping local vulnerability.

Key findings include the spatially varying effectiveness of federal stimulus, with Economic Impact Payments exerting stronger influence in rural areas characterized by tighter economic interconnectedness, while having muted effects in urban and suburban centers.

We also document regional differences in unexplained consumption variation, which are comparatively larger in suburban counties. This pattern may reflect pandemic-related migration flows and labor market reallocation. Notably, regions such as Indiana and Ohio experienced especially severe consumption downturns, suggesting localized economic fragility beyond what national aggregates can reveal.

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